

# Driver Drowsiness Detection

A classical image-processing pipeline that classifies eye state from infrared eye crops, then tracks eyelid closure over time to raise a fatigue alert. No learned feature extractor: every stage is a textbook DIP operator.

89.65 %

TEST ACCURACY

0.962

ROC AUC

46.8 fps

FULL PIPELINE

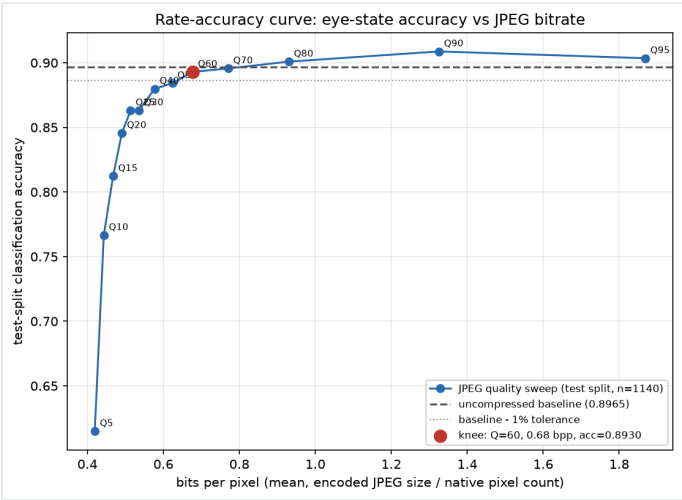
24.3 KB/s

AT COMPRESSION KNEE

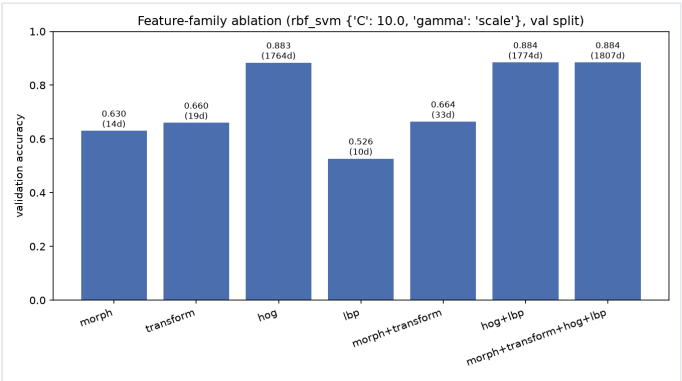
## PIPELINE

|                                                                                                           |                                                                                                     |                                                                                                |                                                                                           |                                                                                              |                                                                                          |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 01<br><b>Enhancement</b><br>CLAHE (clip 2.0, tile 4x4), gamma 1.5, glare inpainting. Normalised to 64x64. | 02<br><b>Filtering</b><br>Median 3x3 for impulse noise, mean 5x5 for Gaussian. Chosen by PSNR/SSIM. | 03<br><b>Segmentation</b><br>Otsu thresholding and seeded region growing to isolate the pupil. | 04<br><b>Morphology</b><br>Opening 3x3, closing 5x5, hole-fill. 14 region-shape features. | 05<br><b>Features</b><br>HOG + LBP texture descriptors, 1774 dimensions, RBF-SVM classifier. | 06<br><b>Temporal</b><br>PERCLOS over a 10 s window with hysteresis to suppress flicker. |
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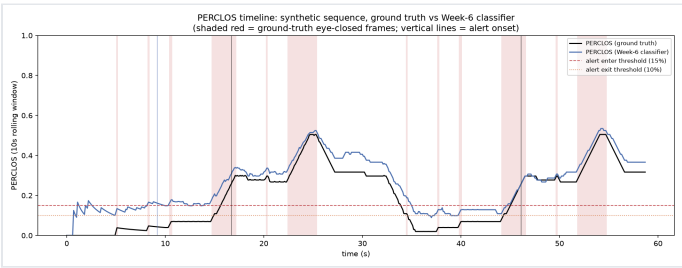
## RESULTS



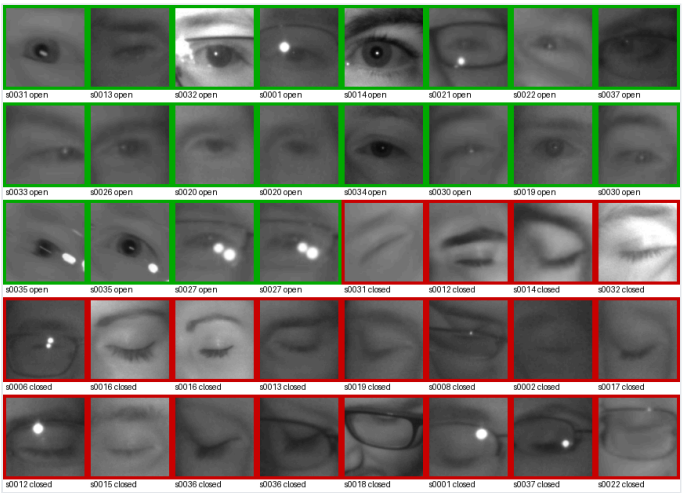
**Rate-accuracy trade-off.** Accuracy holds within 0.35 points of the uncompressed baseline down to JPEG quality 60, or 0.811 KB per frame.



**Feature ablation.** Texture descriptors reach 0.884; adding all segmentation, morphology and transform features changes it by 0.0000.



**PERCLOS timeline.** Per-frame accuracy 0.903 on the validation sequence, but alert-timing agreement only 0.711.



**Dataset.** Infrared eye crops spanning eyeglasses, glare and low light. Splits are subject-disjoint to prevent identity leakage.

## WHAT THE MEASUREMENTS SHOWED

### Texture outperformed shape

Segmentation and transform features carried real signal alone (0.630 and 0.660) but proved redundant once texture descriptors were present — a measured result, not an assumption.

### Quality metrics mislead

At JPEG Q15, SSIM reads 0.91 while classification has already lost 8.4 points. Optimising a vision pipeline for PSNR or SSIM optimises the wrong objective.

### Denoising rescues segmentation

Salt-and-pepper noise drove Otsu mask IoU to 0.000 on a poorly-lit sample; a median 3x3 filter beforehand restored it to 0.90–0.98.